Homework 2

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1. Question 1

1.a.

Yes, see reasoning in 1b.

1.b.

Define the payoff as X, price vector as P, and price of contingent claim as p. Solve the linear equation $Xp = P$ by elimination, we got

$$[X, P] \longrightarrow \begin{bmatrix} 1 & 0 & 0 & 0.2 \\ 0 & 1 & 0 & 0.4 \\ 0 & 0 & 1 & 0.2 \end{bmatrix} \tag{1}$$

Thus, we have complete markets in this economy and the price of continguent claim for state of nature 1, 2,3 are $[0.2, 0.4, 0.2]'$

1.c.

The price should be

$$[30, 20, 10] \times [0.2, 0.4, 0.2]' = 16 \tag{2}$$

1.d.

SDP in this economy is calucalteed by $M(i) = \frac{p(i)}{\pi(i)}$

$$M = [0.8, 0.8, 0.8]'. \tag{3}$$

This economy satisfies no-arbitrage as all the prices of contingent claims are strictly positive and unique. SDF remains the same accross all states, which means that agents in this economy are risk neutral.

1.e.

$$P_D = \sum_i \pi(i) M(i) X(i) = \frac{1}{4} \cdot 0.8 \cdot 30 + \frac{1}{2} \cdot 0.8 \cdot 20 + \frac{1}{4} \cdot 0.8 \cdot 10 = 16 \tag{4}$$

Homework
Solution

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2. Question 2

2.a.

Parameter	Estimate	Std. error	t-stat
a	1.0368	0.0119	86.86
b_1 (Lo 10)	−1.5850	1.6173	−0.98
b_2 (2-Dec)	3.5503	1.8540	1.91
b_3 (3-Dec)	0.7104	2.0766	0.34
b_4 (4-Dec)	−3.1122	1.8756	−1.66
b_5 (5-Dec)	1.1837	2.0043	0.59
b_6 (6-Dec)	2.9972	1.9857	1.51
b_7 (7-Dec)	−5.2931	1.8685	−2.83
b_8 (8-Dec)	2.4722	1.6310	1.52
b_9 (9-Dec)	2.8775	1.6288	1.77
b_{10} (Hi 10)	−1.1771	1.0598	−1.11

Two-step GMM, monthly, 1927:07–2026:08 (T = 1190). Excess returns in decimals. Newey-West S with automatic lag L = 19.

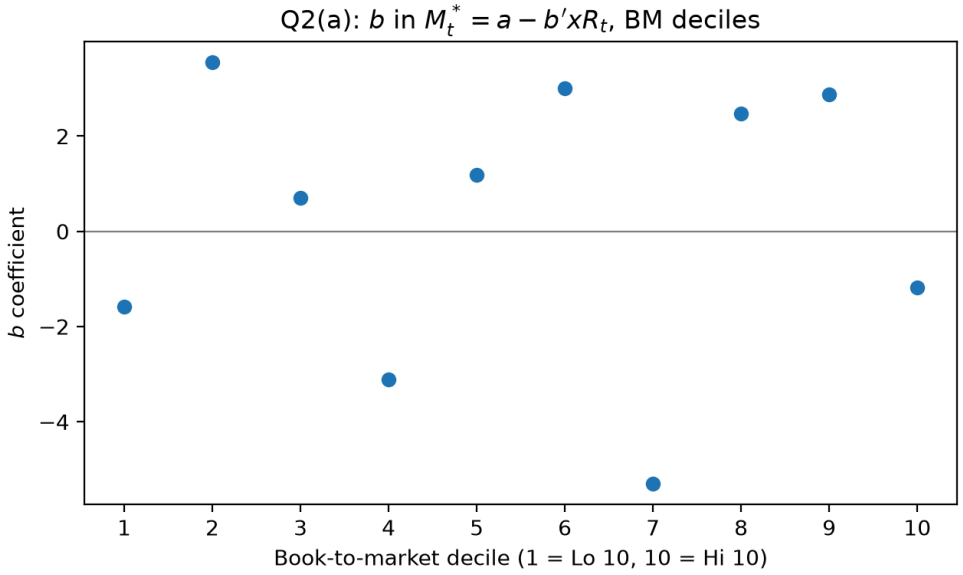


Figure 1: b coefficients of $M_t^* = a - b'xR_t$ by book-to-market decile (two-step GMM, 1927:07–2026:08).

2.b.

Parameter	Estimate	Std. error	t-stat
a	1.0666	0.0154	69.32
b_1 (Lo PRIOR)	−4.0194	0.9747	−4.12

Parameter	Estimate	Std. error	t-stat
b_2 (PRIOR 2)	1.1429	1.4347	0.80
b_3 (PRIOR 3)	0.1024	1.8005	0.06
b_4 (PRIOR 4)	1.4897	1.7155	0.87
b_5 (PRIOR 5)	2.5317	2.0073	1.26
b_6 (PRIOR 6)	−0.8195	1.8827	−0.44
b_7 (PRIOR 7)	0.2777	1.8301	0.15
b_8 (PRIOR 8)	0.9818	1.7812	0.55
b_9 (PRIOR 9)	−0.9414	1.6701	−0.56
b_{10} (Hi PRIOR)	3.4370	0.8792	3.91

Two-step GMM, monthly, 1927:07–2026:08 (T = 1190). Excess returns in decimals. Newey-West S with automatic lag L = 3.

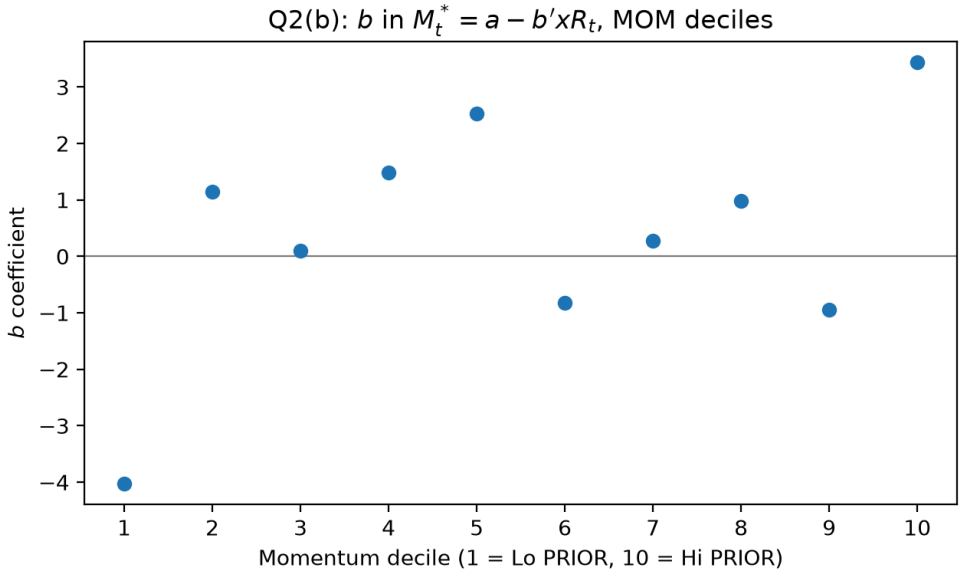


Figure 2: b coefficients of $M_t^* = a - b'xR_t$ by momentum decile (two-step GMM, 1927:07–2026:08).

The sample correlation between M_t^* from Question 2(a) (BM deciles) and M_t^* from Question 2(b) (MOM deciles) is $\mathrm{Corr}\left(M_{BM,t}^*, M_{MOM,t}^*\right) = 0.3555$ (monthly, 1927:07–2026:08, $T = 1190$).

This correlation means that two SDF has 35.55% explanation power pricing same portion of return variation.

2.c.

Parameter	Estimate	Std. error	t-stat
a	1.0927	0.0195	56.14
b_1 (Lo 10)	−1.9664	1.8048	−1.09

Parameter	Estimate	Std. error	t-stat
b_2 (2-Dec)	2.4739	1.8751	1.32
b_3 (3-Dec)	0.5781	2.1303	0.27
b_4 (4-Dec)	−3.7439	1.8387	−2.04
b_5 (5-Dec)	1.0655	2.1028	0.51
b_6 (6-Dec)	2.6840	1.7591	1.53
b_7 (7-Dec)	−4.6456	1.7091	−2.72
b_8 (8-Dec)	2.4707	1.6922	1.46
b_9 (9-Dec)	3.7398	1.6604	2.25
b_{10} (Hi 10)	−0.2415	1.0046	−0.24
b_{11} (Lo PRIOR)	−4.4169	1.1081	−3.99
b_{12} (PRIOR 2)	0.6354	1.5542	0.41
b_{13} (PRIOR 3)	−0.1712	1.8918	−0.09
b_{14} (PRIOR 4)	1.9480	1.8068	1.08
b_{15} (PRIOR 5)	1.4236	2.1083	0.68
b_{16} (PRIOR 6)	−0.8781	2.2007	−0.40
b_{17} (PRIOR 7)	−0.1639	1.8900	−0.09
b_{18} (PRIOR 8)	0.9802	1.8749	0.52
b_{19} (PRIOR 9)	−1.2708	1.8496	−0.69
b_{20} (Hi PRIOR)	3.4473	1.0952	3.15

Two-step GMM, monthly, 1927:07–2026:08 (T = 1190). Excess returns in decimals. Newey-West S with automatic lag L = 7.

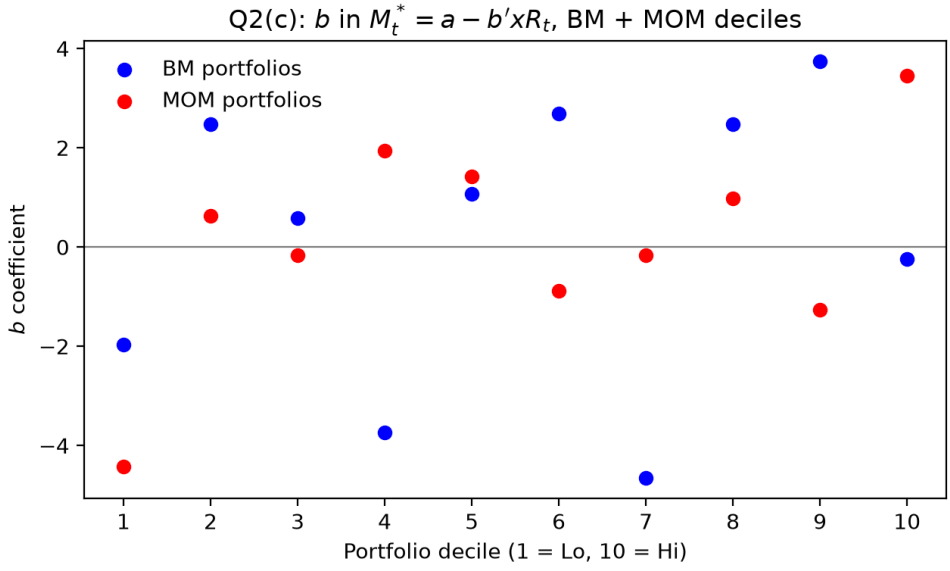


Figure 3: b coefficients of $M_t^* = a - b' x R_t$ estimated on the 20 BM and MOM decile portfolios jointly, by decile (blue: BM portfolios; red: MOM portfolios; two-step GMM, 1927:07–2026:08).

Question	Test assets	Annualized Sharpe ratio of $xR_t^* = b' x R_t$
2(a)	10 BM deciles	0.6646
2(b)	10 MOM deciles	0.8937
2(c)	20 BM + MOM deciles	1.0545

Monthly 1927:07–2026:08 (T = 1190); Sharpe ratio = $\sqrt{12}$ × mean / s.d. of the monthly xR_t^* series.

SDF constructed using 20 portfolios has higher sharpe ratio than other two. Higher sharpe ratio means pricing power is high. As in the linear pricing SDF setting, the highest sharpe ratio portfolio can price every return.

2.d.

Portfolio	$\hat{\alpha}_p$, M^* from 2(a) (BM)	$\hat{\alpha}_p$, M^* from 2(b) (MOM)	$\hat{\alpha}_p$, M^* from 2(c) (BM + MOM)
BM Lo 10	0.0000	−0.1511	0.0000
BM 2-Dec	0.0000	−0.0101	0.0000
BM 3-Dec	0.0000	−0.0223	0.0000
BM 4-Dec	0.0000	−0.0778	0.0000
BM 5-Dec	0.0000	0.0504	0.0000
BM 6-Dec	0.0000	0.1261	0.0000
BM 7-Dec	0.0000	0.0281	0.0000
BM 8-Dec	0.0000	0.2376	0.0000
BM 9-Dec	0.0000	0.3450	0.0000
BM Hi 10	0.0000	0.3198	0.0000
MOM Lo PRIOR	−1.0157	0.0000	0.0000
MOM PRIOR 2	−0.5051	0.0000	0.0000
MOM PRIOR 3	−0.3272	0.0000	0.0000
MOM PRIOR 4	−0.1488	0.0000	0.0000
MOM PRIOR 5	−0.0882	0.0000	0.0000
MOM PRIOR 6	−0.0263	0.0000	0.0000
MOM PRIOR 7	0.0509	0.0000	0.0000
MOM PRIOR 8	0.1902	0.0000	0.0000
MOM PRIOR 9	0.2240	0.0000	0.0000
MOM Hi PRIOR	0.5418	0.0000	0.0000

Pricing errors $\hat{\alpha}_p = \hat{E}\left[xR_p\right] - \widehat{Cov}\left[-M^*, xR_p\right]$ in percent per month; monthly 1927:07–2026:08 (T = 1190); covariance with divisor T.

Every SDF perfectly prices excess returns of portfolios where SDF comes from, as in each case it is just identified. However, SDF from some portfolios cannot perfectly price returns of other portfios. We can increase the portfolio number to increase pricing ability.

2.e.

Estimates of $M_t^* = a - b'xR_t$ on the 20 BM and MOM decile portfolios using data until December 1989:

Parameter	Estimate	Std. error	t-stat
a	1.1248	0.0291	38.59
b_1 (Lo 10)	−3.3377	2.0475	−1.63
b_2 (2-Dec)	4.1839	2.4198	1.73
b_3 (3-Dec)	−1.1689	3.3312	−0.35
b_4 (4-Dec)	−1.7978	2.2125	−0.81
b_5 (5-Dec)	−0.0806	2.7866	−0.03
b_6 (6-Dec)	1.8808	1.8149	1.04
b_7 (7-Dec)	−4.1249	2.2798	−1.81
b_8 (8-Dec)	4.9617	1.9088	2.60
b_9 (9-Dec)	3.7239	1.7576	2.12
b_{10} (Hi 10)	0.1428	1.2633	0.11
b_{11} (Lo PRIOR)	−7.1059	1.7249	−4.12
b_{12} (PRIOR 2)	1.8777	1.8779	1.00
b_{13} (PRIOR 3)	−0.5889	2.8536	−0.21
b_{14} (PRIOR 4)	0.7084	1.9264	0.37
b_{15} (PRIOR 5)	2.2746	3.1138	0.73
b_{16} (PRIOR 6)	−0.8297	2.8608	−0.29
b_{17} (PRIOR 7)	−0.9012	3.0357	−0.30
b_{18} (PRIOR 8)	−0.1210	2.1700	−0.06
b_{19} (PRIOR 9)	0.4871	2.0593	0.24
b_{20} (Hi PRIOR)	3.2646	1.6282	2.00

Two-step GMM, monthly, 1927:07–1989:12 (T = 750). Excess returns in decimals. Newey-West S with automatic lag L = 21.

Portfolio	$\hat{\alpha}_{OS,p}$
BM Lo 10	1.0971
BM 2-Dec	0.7858
BM 3-Dec	0.9002
BM 4-Dec	0.6957
BM 5-Dec	0.7986

Portfolio	$\hat{\alpha}_{OS,p}$
BM 6-Dec	0.7178
BM 7-Dec	0.6977
BM 8-Dec	0.5923
BM 9-Dec	0.8269
BM Hi 10	0.9709
MOM Lo PRIOR	2.7055
MOM PRIOR 2	1.6483
MOM PRIOR 3	1.3905
MOM PRIOR 4	1.2077
MOM PRIOR 5	0.9180
MOM PRIOR 6	0.8297
MOM PRIOR 7	0.7601
MOM PRIOR 8	0.6444
MOM PRIOR 9	0.4322
MOM Hi PRIOR	0.5453

Out-of-sample pricing errors $\hat{\alpha}_{OS,p} = \hat{E}\left[xR_p\right] - \widehat{Cov}\left[-M^*_{OS}, xR_p\right]$ in percent per month, computed over 1990:01–2026:08 (T = 440), covariance with divisor T. $M^*_{OS,t} = \hat{a} - \hat{b}' xR_t$ with $\hat{a}, \hat{b}$ estimated over 1927:07–1989:12.

Out of sample pricing errors are non-zero for all portfolios, which means the true values for a, b are time varying.

3. Question 3

3.a.

The portfolio choice problem is

$$\max_{\bar{\omega}_t} V = \frac{1}{1-\gamma} E\Big[\left(W_t R_{f,t+1} + \bar{\omega}_t{}' W_t xR_{t+1}\right)^{1-\gamma}\Big] \tag{5}$$

FOC w.r.t $\bar{\omega}_t$,

$$\frac{\partial}{\partial \bar{\omega}_t} V = E\Big[\left(R_{f,t+1} + \bar{\omega}_t{}' xR_{t+1}\right)^{-\gamma} W_t^{1-\gamma} xR_{t+1}\Big] = 0 \tag{6}$$

We can put $W_t^{1-\gamma}$ outside of expectation as it is known at time t and divide it from both sides. Then the FOC becomes

$$E\Big[R^{-\gamma}_{\omega^*_t,t+1} xR_{t+1}\Big] = E\Big[\left(R_{f,t+1} + \bar{\omega}^*_t{}' xR_{t+1}\right)^{-\gamma} xR_{t+1}\Big] = 0. \tag{7}$$

Thus, $M_t = R^{-\gamma}_{\omega^*_t,t+1}$ prices all excess returns.

3.b.

Parameter	Estimate	Std. error	t-stat

Parameter	Estimate	Std. error	t-stat
γ	1.9061	0.6026	3.16

GMM with the single moment $\hat{E}\left[R_{w,t}^{-\gamma} x R_{w,t}\right] = 0$, $R_{w,t} = e^{r_{e,t}}$, $xR_{w,t} = e^{r_{e,t}} - e^{r_{f,t}}$; monthly observations of annual returns, 1927:12–2021:12 (T = 1129). Newey-West (1987, 1994) S with automatic lag L = 26.

3.c.

	Estimate	Std. error	t-stat
γ	3.0147	0.7174	4.20
Pricing error: $\hat{E}\left[R_{w,t}^{-\gamma} x R_{w,t}\right]$	−0.00407	0.00464	−0.88
Pricing error: $\hat{E}\left[R_{w,t}^{-\gamma} z_{t-1} x R_{w,t}\right]$	0.00194	0.00221	0.88

χ^2 test that the pricing errors are jointly zero: $\chi^2(1) = 0.770$, p-value = 0.3803.

GMM with identity weighting matrix; monthly observations of annual returns, 1940:12–2021:12 (T = 973). Newey-West (1987, 1994) S (centered) with automatic lag L = 24.

3.d.

	Estimate	Std. error	t-stat
γ	2.9736	0.7109	4.18
Pricing error: $\hat{E}\left[R_{w,t}^{-\gamma} x R_{w,t}\right]$	−0.00275	0.00309	−0.89
Pricing error: $\hat{E}\left[R_{w,t}^{-\gamma} z_{t-1} x R_{w,t}\right]$	0.00472	0.00532	0.89

χ^2 test that the pricing errors are jointly zero: $\chi^2(1) = 0.789$, p-value = 0.3743.

Two-step GMM, second step weighted by the inverse of the Q3(c) S (lag 24); monthly observations of annual returns, 1940:12–2021:12 (T = 973). Newey-West (1987, 1994) S (centered) with automatic lag L = 24.

3.e.

	Estimate	Std. error	t-stat
γ	2.9999	0.7044	4.26
Pricing error: $\hat{E}\left[R_{w,t}^{-\gamma} x R_{w,t}\right]$	−0.00359	0.00410	−0.88
Pricing error: $\hat{E}\left[R_{w,t}^{-\gamma} z_{t-1} x R_{w,t}\right]$	0.00294	0.01001	0.29
Pricing error: $\hat{E}\left[R_{w,t}^{-\gamma} x R_{p,t}\right]$, BM Lo 10	−0.01758	0.01152	−1.53
Pricing error: $\hat{E}\left[R_{w,t}^{-\gamma} x R_{p,t}\right]$, BM 2-Dec	0.00118	0.00733	0.16
Pricing error: $\hat{E}\left[R_{w,t}^{-\gamma} x R_{p,t}\right]$, BM 3-Dec	−0.00402	0.00608	−0.66
Pricing error: $\hat{E}\left[R_{w,t}^{-\gamma} x R_{p,t}\right]$, BM 4-Dec	0.00022	0.00603	0.04
Pricing error: $\hat{E}\left[R_{w,t}^{-\gamma} x R_{p,t}\right]$, BM 5-Dec	0.01488	0.00687	2.17
Pricing error: $\hat{E}\left[R_{w,t}^{-\gamma} x R_{p,t}\right]$, BM 6-Dec	0.02023	0.00760	2.66
Pricing error: $\hat{E}\left[R_{w,t}^{-\gamma} x R_{p,t}\right]$, BM 7-Dec	0.00559	0.01186	0.47

Pricing error: $\hat{E}\left[R_{w,t}^{-\gamma} x R_{p,t}\right]$, BM 8-Dec	0.02965	0.00989	3.00
	Estimate	Std. error	t-stat
Pricing error: $\hat{E}\left[R_{w,t}^{-\gamma} x R_{p,t}\right]$, BM 9-Dec	0.03987	0.01038	3.84
Pricing error: $\hat{E}\left[R_{w,t}^{-\gamma} x R_{p,t}\right]$, BM Hi 10	0.03322	0.01561	2.13
Pricing error: $\hat{E}\left[R_{w,t}^{-\gamma} x R_{p,t}\right]$, MOM Lo PRIOR	−0.10521	0.01384	−7.60
Pricing error: $\hat{E}\left[R_{w,t}^{-\gamma} x R_{p,t}\right]$, MOM PRIOR 2	−0.04392	0.00922	−4.76
Pricing error: $\hat{E}\left[R_{w,t}^{-\gamma} x R_{p,t}\right]$, MOM PRIOR 3	−0.02020	0.00894	−2.26
Pricing error: $\hat{E}\left[R_{w,t}^{-\gamma} x R_{p,t}\right]$, MOM PRIOR 4	−0.00272	0.00703	−0.39
Pricing error: $\hat{E}\left[R_{w,t}^{-\gamma} x R_{p,t}\right]$, MOM PRIOR 5	0.00105	0.00733	0.14
Pricing error: $\hat{E}\left[R_{w,t}^{-\gamma} x R_{p,t}\right]$, MOM PRIOR 6	0.00161	0.00569	0.28
Pricing error: $\hat{E}\left[R_{w,t}^{-\gamma} x R_{p,t}\right]$, MOM PRIOR 7	0.00900	0.00768	1.17
Pricing error: $\hat{E}\left[R_{w,t}^{-\gamma} x R_{p,t}\right]$, MOM PRIOR 8	0.01968	0.00540	3.65
Pricing error: $\hat{E}\left[R_{w,t}^{-\gamma} x R_{p,t}\right]$, MOM PRIOR 9	0.02240	0.00680	3.29
Pricing error: $\hat{E}\left[R_{w,t}^{-\gamma} x R_{p,t}\right]$, MOM Hi PRIOR	0.04974	0.01125	4.42

χ^2 test that the pricing errors are jointly zero: $\chi^2(21) = 217.092$, p-value < 0.0001.

GMM with identity weighting matrix; BM/MOM excess returns are 12-month compounded nominal returns minus the 12-month compounded RF (Ken French); monthly observations of annual returns, 1940:12–2021:12 (T = 973). Newey-West (1987, 1994) S (centered) with automatic lag L = 24.